

YEAR 7 MATHEMATICS

Chapter 11

Probability

WORKSHEETS

11

How to use this booklet

1 There is **one worksheet for each lesson**. It has three columns that get harder from left to right: **Easy**, then **Medium**, then **Challenging**.

WE DO **Top of each column.** Your teacher works the examples on the board. **Copy every step** into the space.

2 Read the **speech bubble** at the top of each column: it tells you what to do. A new bubble means a new kind of question.

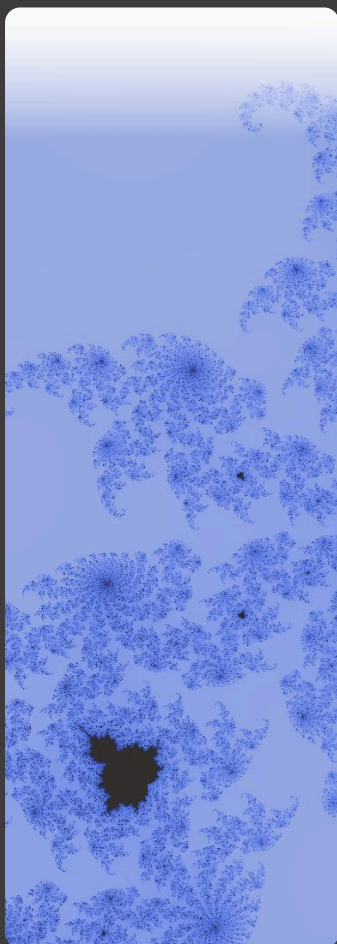
YOU DO **The rest of the column.** Your turn. Column 1: answer on the line. Columns 2 and 3: show your working on the lines.

Contents and progress tracker

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In this chapter you will

- list the sample space of an experiment
- find the probability of simple events as fractions, decimals and percentages
- use the probability scale and chance words
- compare experimental and theoretical probability
- find the probability of complementary events



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is feathery spirals deep in Elephant Valley. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

Easy

Questions 1–13

How many outcomes are in the sample space?



WE DO With your teacher: copy the working.

E1 tossing a coin _____

E2 rolling a die _____

YOU DO Your turn.

1 choosing a day of the week 7

2 choosing a month _____

3 choosing a card from a deck _____

4 choosing a letter of the alphabet _____

5 a spinner with 5 equal sectors _____

6 choosing a suit of cards _____

7 choosing a vowel _____

8 choosing a digit 0–9 _____

9 choosing a letter from MATHS _____

10 choosing a season _____

11 rolling an 8-sided die _____

12 choosing an even number 1–10 _____

13 choosing a colour of the rainbow _____

Medium

Questions 14–21

List the sample space, then count the outcomes.

WE DO With your teacher: copy the working.

E1 rolling a die _____

E2 choosing a letter from CAT _____

YOU DO Your turn.

14 choosing a suit
{ hearts, diamonds, clubs, spades }

4 outcomes

15 an odd number less than 10
{ 1, 3, 5, 7, 9 }
____ outcomes

16 a weekend day

____ outcomes

17 a vowel

18 a prime less than 12

19 a letter from BOOK

20 a multiple of 5 up to 30

21 a factor of 12

Challenging

Questions 22–26

Two things happen together. Multiply to count the outcomes, then list a few.



WE DO With your teacher: copy the working.

E1 toss a coin and roll a die _____

E2 toss two coins _____

YOU DO Your turn.

22 roll two dice
 $6 \times 6 = 36$
e.g. (____, ____), (____, ____), ..., (____, ____)

23 choose a shirt (3) and pants (4)
____ × ____ = ____
e.g. S____P____, S____P____, ...

24 spin a 4-colour spinner and toss a coin

25 choose a day of the week and toss a coin

26 choose an entrée (2) and a main (5)

Easy

Questions 1–12

A fair die is rolled. Find the probability as a fraction.



WE DO With your teacher: copy the working.

E1 P(a 4) _____

E2 P(an even number) _____

YOU DO Your turn.

1 P(an odd number) $\frac{1}{2}$

2 P(more than 4) _____

3 P(less than 3) _____

4 P(a 7) _____

5 P(a multiple of 3) _____

6 P(a prime) _____

7 P(not a 6) _____

8 P(at least 5) _____

9 P(a factor of 6) _____

10 P(a square number) _____

11 P(less than 10) _____

12 P(a 1 or a 2) _____

Medium

Questions 13–19

Count the favourable outcomes and the total, then simplify.



WE DO With your teacher: copy the working.

E1 A bag has 3 green and 9 red counters. P(green) _____

E2 A bag has 1 green and 5 yellow counters. P(green) _____

YOU DO Your turn.

13 A bag has 3 green and 4 red counters. P(green)

3 green out of 7

$$P = \frac{3}{7} = \frac{3}{7}$$

14 A bag has 3 blue and 4 green counters. P(blue)

3 blue out of 7

$$P = \frac{3}{7} = \frac{3}{7}$$

15 A bag has 8 green and 1 yellow counters. P(green)

___ green out of ___

$$P = \frac{8}{9} = \frac{8}{9}$$

16 A bag has 8 yellow and 2 blue counters. P(yellow)

17 A bag has 2 red and 6 green counters. P(red)

18 A bag has 7 yellow and 5 red counters. P(yellow)

19 A bag has 9 blue and 9 red counters. P(blue)

Challenging

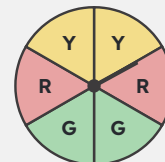
Questions 20–21

Use the spinner (R = red, B = blue, G = green, Y = yellow). Each sector is the same size.



WE DO With your teacher: copy the working.

E1 P(green), as a fraction and a percentage



E2 P(yellow), as a fraction and a percentage



YOU DO Your turn.

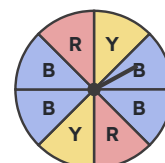
20 P(green), as a fraction and a percentage



2 of the 6 sectors are green

$$P = \frac{2}{6} = \frac{1}{3} = \frac{33}{100} \%$$

21 P(yellow), as a fraction and a percentage



___ of the ___ sectors are yellow

$$P = \frac{2}{8} = \frac{1}{4} = \frac{25}{100} \%$$

11.03 The probability scale

YEAR 7 · CHAPTER 11: PROBABILITY

Easy

Questions 1–28

Write the chance word: impossible, unlikely, even chance, likely or certain.



WE DO

With your teacher: copy the working.

E1 0.75

E2 5%

YOU DO

Your turn.

1 0.95

2 30%

3 40%

4 0.4

5 0.3

6 100%

7 50%

8 0.2

9 70%

10 25%

11 60%

12 20%

13 0.1

14 90%

15 0

16 95%

17 0.25

18 0.8

19 80%

20 0.9

21 0.6

22 0.05

23 0.5

24 0.7

25 1

26 0%

27 10%

28 75%

likely

Medium

Questions 29–35

Write the probability as a fraction, then choose the chance word.



WE DO

With your teacher: copy the working.

E1 2 red and 5 blue counters: P(red)

E2 0 red and 6 blue counters: P(red)

YOU DO

Your turn.

29 4 red and 5 blue counters: P(red)

$$P(\text{red}) = \frac{4}{9} = \frac{4}{9}$$

unlikely

30 3 red and 1 blue counters: P(red)

$$P(\text{red}) = \frac{3}{4} = \frac{3}{4}$$

31 0 red and 5 blue counters: P(red)

$$P(\text{red}) = \frac{0}{5} = \frac{0}{5}$$

32 4 red and 6 blue counters: P(red)

33 1 red and 0 blue counters: P(red)

34 1 red and 4 blue counters: P(red)

35 0 red and 1 blue counters: P(red)

Challenging

Questions 36–43

Which is more likely? Change both to decimals first.



WE DO

With your teacher: copy the working.

E1 $\frac{2}{5}$ or 55%

E2 $\frac{3}{8}$ or 45%

YOU DO

Your turn.

36 $\frac{3}{8}$ or 15%

$$\frac{3}{8} = 0.375, 15\% = 0.15$$

more likely: $\frac{3}{8}$

37 $\frac{3}{8}$ or 70%

$$\frac{3}{8} = \frac{3}{8}, 70\% = 0.7$$

more likely: 70%

38 $\frac{7}{10}$ or 30%

39 $\frac{7}{10}$ or 65%

40 $\frac{5}{8}$ or 65%

41 $\frac{1}{4}$ or 35%

42 $\frac{2}{3}$ or 55%

43 $\frac{2}{5}$ or 35%

Easy

Questions 1–21

Find the expected number:
probability $\times$ number of trials.



WE DO

With your teacher: copy the working.

- E1 heads in 36 trials _____
- E2 a 1 or 2 on a die in 48 trials _____

YOU DO

Your turn.

- 1 a 1 or 2 on a die in 84 trials 28
- 2 an even number on a die in 12 trials _____
- 3 an even number on a die in 20 trials _____
- 4 a 1 or 2 on a die in 81 trials _____
- 5 heads in 54 trials _____
- 6 a 6 on a die in 24 trials _____
- 7 an even number on a die in 6 trials _____
- 8 an even number on a die in 16 trials _____
- 9 a heart in 64 trials _____
- 10 a 6 on a die in 54 trials _____
- 11 a 6 on a die in 150 trials _____
- 12 a 1 or 2 on a die in 57 trials _____
- 13 a 6 on a die in 18 trials _____
- 14 a 6 on a die in 90 trials _____
- 15 a 1 or 2 on a die in 75 trials _____
- 16 a 1 or 2 on a die in 27 trials _____
- 17 heads in 26 trials _____
- 18 a heart in 72 trials _____
- 19 heads in 56 trials _____
- 20 a 6 on a die in 36 trials _____
- 21 heads in 44 trials _____

Medium

Questions 22–29

Find the experimental probability: times it happened $\div$ trials. Simplify.



WE DO

With your teacher: copy the working.

- E1 25 heads in 50 tosses _____
- E2 3 heads in 50 tosses _____

YOU DO

Your turn.

- 22 38 heads in 100 tosses
 $\frac{38}{100}$
 $= \frac{19}{50}$
- 23 19 heads in 60 tosses
 $\frac{19}{60}$
 $= \frac{19}{60}$
- 24 17 heads in 50 tosses
 $\frac{17}{50}$
 $= \frac{17}{50}$
- 25 2 heads in 20 tosses _____
- 26 86 heads in 100 tosses _____
- 27 57 heads in 60 tosses _____
- 28 31 heads in 100 tosses _____
- 29 23 heads in 40 tosses _____

Challenging

Questions 30–35

Compare the experiment with the theory. Show both probabilities as percentages.



WE DO

With your teacher: copy the working.

- E1 A 6 came up 25 times in 120 rolls. Is the die fair?

- E2 A 6 came up 9 times in 30 rolls. Is the die fair?

YOU DO

Your turn.

- 30 A 6 came up 16 times in 60 rolls. Is the die fair?
experiment: $16 \div 60 = 26.7\%$
theory: $1 \div 6 = 16.7\%$
not close: _____
- 31 A 6 came up 20 times in 120 rolls. Is the die fair?
experiment: $\frac{\quad}{\quad} = \frac{\quad}{\quad}\%$
theory: $\frac{\quad}{\quad} = \frac{\quad}{\quad}\%$

- 32 A 6 came up 18 times in 120 rolls. Is the die fair?

- 33 A 6 came up 12 times in 60 rolls. Is the die fair?

- 34 A 6 came up 10 times in 60 rolls. Is the die fair?

- 35 A 6 came up 7 times in 30 rolls. Is the die fair?

11.05 Complementary events

YEAR 7 · CHAPTER 11: PROBABILITY

Easy

Questions 1–30

Find $P(\text{not } E) = 1 - P(E)$.

WE DO With your teacher: copy the working.

E1 $P(E) = 0.38$

E2 $P(E) = 0.2$

YOU DO Your turn.

1 $P(E) = 0.67$

0.33

2 $P(E) = 0.75$

3 $P(E) = \frac{2}{3}$

4 $P(E) = \frac{2}{9}$

5 $P(E) = 0.92$

6 $P(E) = \frac{1}{2}$

7 $P(E) = \frac{1}{6}$

8 $P(E) = 0.11$

9 $P(E) = \frac{2}{5}$

10 $P(E) = \frac{11}{12}$

11 $P(E) = \frac{1}{8}$

12 $P(E) = 0.86$

13 $P(E) = 0.25$

14 $P(E) = 0.22$

15 $P(E) = \frac{1}{4}$

16 $P(E) = \frac{5}{6}$

17 $P(E) = 0.8$

18 $P(E) = \frac{3}{8}$

19 $P(E) = 0.29$

20 $P(E) = 0.14$

21 $P(E) = \frac{3}{10}$

22 $P(E) = \frac{1}{7}$

23 $P(E) = \frac{1}{10}$

24 $P(E) = \frac{1}{5}$

25 $P(E) = \frac{8}{9}$

26 $P(E) = \frac{5}{7}$

27 $P(E) = \frac{4}{9}$

28 $P(E) = 0.88$

29 $P(E) = 0.5$

30 $P(E) = \frac{9}{10}$

Medium

Questions 31–35

Describe the complementary event, then find its probability.

WE DO With your teacher: copy the working.

E1 rolling a 6

E2 tossing heads

YOU DO Your turn.

31 a red card

complement: a black card

$$P = 1 - \frac{1}{2} = \frac{1}{2}$$

32 an even number on a die

complement: an odd number

$$P = \frac{1}{6} - \frac{1}{6} = \frac{1}{6}$$

33 a heart

complement: _____

$$P = \frac{1}{4} - \frac{1}{4} = \frac{3}{4}$$

34 rolling less than 3

35 a vowel from MATHS

Challenging

Questions 36–40

Find $P(\text{not the colour given})$. Find the total first.

WE DO With your teacher: copy the working.

E1 9 red, 9 blue, 9 green: $P(\text{not green})$

E2 3 red, 1 blue, 7 green: $P(\text{not blue})$

YOU DO Your turn.

36 8 red, 2 blue, 6 green: $P(\text{not red})$

total: 16

not red: $16 - 8 = 8$

$$P = \frac{8}{16} = \frac{1}{2}$$

37 4 red, 3 blue, 7 green: $P(\text{not red})$

total: ____

not red: ____ - ____ = ____

$$P = \frac{10}{17} = \frac{5}{8.5}$$

38 7 red, 4 blue, 2 green: $P(\text{not green})$

39 9 red, 7 blue, 5 green: $P(\text{not red})$

40 2 red, 2 blue, 1 green: $P(\text{not blue})$